



WEST BENGAL STATE UNIVERSITY
B.Sc. Minor/Cor 2nd Semester Examination, 2024

MTMMIN202T/MTMCOR202T-MATHEMATICS

Time Allotted: 2 Hours

Full Marks: 50

*The figures in the margin indicate full marks.
Candidates should answer in their own words and adhere to the word limit as practicable.
All symbols are of usual significance.*

Answer Question No. 1 and any five from the rest

1. Answer any **five** questions from the following:

2×5 = 10

(a) If $u = \frac{x^2 + y^2}{\sqrt{x+y}}$, $(x, y) \neq (0, 0)$, what should be the value of k so that

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = ku?$$

(b) Find $f'(0)$, if exists, for $f(x) = \begin{cases} 3+2x, & \text{for } -\frac{3}{2} < x \leq 0 \\ 3-2x, & \text{for } 0 < x < \frac{3}{2} \end{cases}$

(c) Is Lagrange's mean value theorem applicable to the function

$$f(x) = \begin{cases} x \sin \frac{1}{x}, & \text{when } x \neq 0 \\ 0, & \text{when } x = 0 \end{cases}$$

in $[-1, 1]$?

(d) Evaluate $\lim_{x \rightarrow 0} x \tan\left(\frac{\pi}{2} - x\right)$.

(e) Show that $\sin x(1 + \cos x)$ has maximum value at $x = \frac{\pi}{3}$.

(f) Find the radius of curvature of $y = e^x$ at $(0, 1)$.

(g) Evaluate: $\int \frac{\sin(2+3 \log x)}{x} dx$

(h) Evaluate: $\int_0^{\pi/2} \sin^4 x \cos^2 x dx$

2. (a) If a function f has a finite derivative at a point, then show that f is continuous at that point. Show by an example that its converse may not be true.

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(b) Find y_4 where $y = e^x \cdot x^3$.

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3. (a) If $y = (\sin^{-1} x)^2$, then show that

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(i) $(1-x^2)y_2 - xy_1 - 2 = 0$

(ii) $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - n^2y_n = 0$

(b) If $u = \frac{x^2 y^2}{x+y}$, applying Euler's Theorem find the value of $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y}$ and hence

deduce that $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = 6u$ (assume that $\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}$).

4. (a) Prove that Rolle's theorem is applicable to the function $f(x) = x^3 - 3x^2 + 2x$ on the interval $[0, 2]$. Find the value of C in $(0, 2)$ that satisfy the theorem and explain why they fulfil the conditions of Rolle's Theorem.

(b) Using Mean Value theorem, show that $\frac{x}{1+x^2} < \tan^{-1} x < x$, for $x > 0$.

5. (a) Evaluate: $\lim_{x \rightarrow 0} (\frac{1}{x^2} - \operatorname{cosec}^2 x)$

(b) Find the Taylor series expansion of $f(x) = \log_e(1+x)$ about $x = 0$.

6. (a) Find the reduction formula for $\int \cos^n x \, dx$.

(b) Prove that $\int_0^\infty e^{-x^2} x^9 \, dx = 12$.

7. (a) Evaluate: $\int_0^1 \frac{dx}{(1-x^6)^{1/6}}$

(b) Prove that $\Gamma(1/2) = \sqrt{\pi}$ (Γ - Gamma).

8. (a) Find the envelopes of the lines $\frac{x}{a} + \frac{y}{b} = 1$ where $a+b=c$, where a and b are parameters and c is constant.

(b) Find the points on the curve $y = 2x^3 - 15x^2 + 34x - 20$ where the tangents are parallel to the straight line $y + 2x = 0$.

9. (a) Find the curvature κ of the curve $y = \log(\cos x)$ at the point where $x = 0$.

(b) Find the asymptotes (if any) of the curve $x^4 - y^4 + 3x^2 y + 3xy^2 + xy = 0$.

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